ASSIGNMENT 3 · MODEL TRAINING & FINE-TUNING · RESEARCH REPORT

Better model or better information? Boosting, fine-tuned encoders and a QLoRA-tuned LLM for predicting failed procurement contracts

Aziz Abdulin · October 2026 · Code & notebooks: github.com/awzwz/tender-risk-ml (notebooks/04–07)

1. Problem and setup

The project predicts, at the moment the winner of a public procurement in Kazakhstan is chosen, whether the resulting contract will **fail** — be terminated or never signed — so that oversight bodies can rank procurements for audit. The data are all 358,789 procurements published in Q1 2024 on the national portal (OWS v3 API) that produced a contract; 6.6 % of them failed. The split is chronological: train = January–February, validation = 1–15 March (all choices), test = 16–31 March (scored once). The selection metric is **PR-AUC** (average precision), because an auditor uses the top of a ranked list; we also report ROC-AUC, F1 at a threshold chosen on validation, precision among the 1 % riskiest (*P@1%*), recall in the top 5 % and the Brier score, and every test difference comes with a paired bootstrap 95 % interval (1,000 resamples). The midterm baseline — a class-weighted logistic regression on 20 features — reached PR-AUC **0.262**. Assignment 3 asks how far this can be pushed and, more importantly, *where* the improvement comes from: a better model, its tuning, or new information about each procurement.

2. Methods

- **Model comparison and tuning (notebook 04).** Logistic regression, random forest, LightGBM (Ke et al., 2017) and CatBoost (Prokhorenkova et al., 2018) on the same 20 features. Logistic regression was tuned by exhaustive grid search (42 configurations, one predefined train → validation fold); the other three by Bayesian optimisation with Optuna/TPE (Akiba et al., 2019; 30 / 50 / 30 trials, early stopping on validation PR-AUC for boosting). Random-fold cross-validation was rejected because it mixes time periods. Separately we measured the effect of the decision threshold and of probability calibration (Platt scaling, isotonic regression).
- **Transfer learning and fine-tuning on lot names (05a/05b).** Three frozen Russian encoders (rubert-tiny2 29 M, multilingual-e5-small 118 M, rubert-base-cased 178 M parameters) were used as feature extractors with a linear probe and, as 32 principal components, inside the tabular model. rubert-tiny2 was fully fine-tuned (all 29.2 M weights; AdamW, weight decay 0.01; learning rate searched in {2e-5, 5e-5, 1e-4}; linear warm-up 6 %; batch 64; 3 epochs with the best epoch chosen on validation; class-weighted cross-entropy; ≤ 20 rows per unique text; three seeds). Out-of-fold scores (5 folds grouped by text) allowed honest stacking. A TF-IDF model served as the no-transfer reference.
- **LLM with QLoRA (06).** Qwen2.5-1.5B read a short Russian "card" of each procurement containing the same facts as the final tabular model plus the lot names, with a two-class classification head. Base weights were frozen in 4-bit NF4 (double quantisation, fp16 compute; Dettmers et al., 2023); LoRA adapters (Hu et al., 2022) with r = 16, α = 32, dropout 0.05 on all seven linear projections (q, k, v, o, gate, up, down) and the full head were trained: **18,467,840 trainable parameters (1.2 %)**. Paged 8-bit AdamW, learning rate 2e-4 with cosine decay, effective batch 64, one epoch on 40,000 train procurements (1 : 3 failure sample), on a free Kaggle T4 (2.2 h training, 1.3 h inference). Full fine-tuning would need ≈ 25 GB without activations; QLoRA peaked at 2.7 GB. Controls: LightGBM on the same 40,000 rows and information, and on the full train set.
- **Final model (07).** 61 features: the 20 midterm features, 32 text components and 9 leak-free versions of rule indicators from the earlier Tender Radar prototype — red flags in the sense of Fazekas & Kocsis (2020) — (customer–supplier history, planned unit price against the market), all computed only from information dated before publication. Three candidates were compared on validation; ablations, SHAP (exact TreeSHAP; Lundberg et al., 2020) and per-segment lift explain the result.

3. Results

3.1 Model family vs tuning

20 midterm features, test period	default	tuned	tuning gain [95 % CI]	what mattered in the search
Logistic regression	0.262	0.280	+0.018 [+0.015, +0.022]	dropping class weights (L1, C = 0.1)
Random forest	0.249	0.290	+0.041 [+0.033, +0.049]	features per split (86 % of variance), depth 13
LightGBM	0.303	0.304	+0.001 [−0.007, +0.010]	learning rate only; early stopping keeps ~9 trees
CatBoost	**0.288**	**0.311**	**+0.023 [+0.016, +0.030]**	**depth 8, no class weights**

Changing the model family is the largest gain on fixed features: tuned CatBoost reaches 0.311 against 0.262 (+0.049 [+0.039, +0.059]); the two boosting libraries end within 0.007 of each other. Tuning helps when the defaults are poor (fully grown random-forest trees overfit and are *below* the linear baseline) and gives nothing for LightGBM, whose validation PR-AUC peaks after a handful of trees — the search ends up choosing where to stop. Convergence curves flatten after about 15 trials. The F1-optimal threshold chosen on validation raises test recall from 5 % (threshold 0.5) to 41 % at 34 % precision; Platt scaling cuts the Brier score of the class-weighted baseline from 0.200 to 0.055 without changing its ranking, while the tuned CatBoost — trained without class weights — is already calibrated.

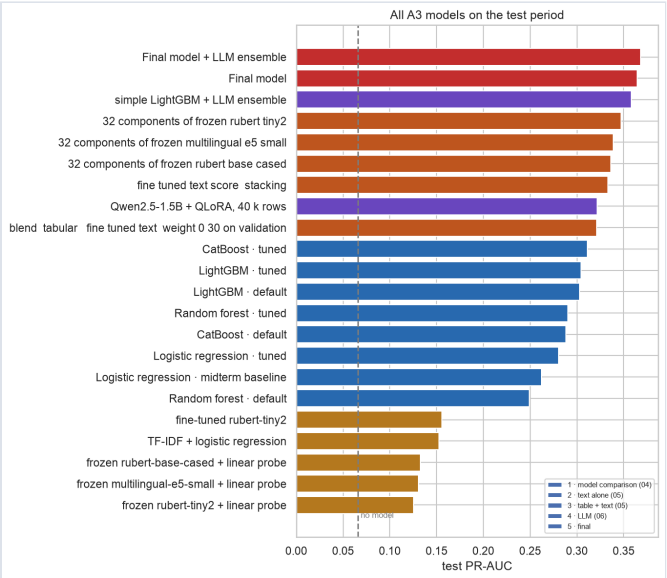
3.2 The lot text: transfer learning vs fine-tuning

Text of the lot names	alone, test	alone, unseen texts	added to 20-feature LightGBM (0.311)
TF-IDF + logistic regression (no transfer)	0.153	0.200	—
Frozen rubert-tiny2 / e5-small / rubert-base + linear probe	0.125 / 0.131 / 0.133	0.172 / 0.189 / 0.187	+0.036 / +0.028 / +0.025 (32 components)
Fine-tuned rubert-tiny2 (mean of 3 seeds)	0.154 ± 0.002	0.232 ± 0.005	+0.022 (stacking), +0.010 (blend)

Frozen encoders transfer general language knowledge (well above the 0.066 base rate) but alone lose to TF-IDF: similarity in general Russian is not procurement risk. Fine-tuning produces the best text model on names never seen in training (0.228–0.237 vs 0.200), so it learns task-specific vocabulary. Inside the tabular model, however, the 32 frozen rubert-tiny2 components are the best way to add text (+0.036 [+0.025, +0.048]): they tell the trees *what* is bought, whereas a single fine-tuned risk score discards that. Larger encoders brought no benefit for five-word catalogue names.

3.3 The LLM

The QLoRA-tuned Qwen2.5-1.5B reaches test PR-AUC **0.322** (ROC-AUC 0.808) — above every model on the 20 features — but below LightGBM given the same 40,000 rows and information — as expected from TabLLM, where the advantage of LLMs on tabular data disappears with many examples (Hegselmann et al., 2023) (0.334; difference −0.011 [−0.024, 0.000]) and further below LightGBM on the full train set (0.350; −0.028 [−0.041, −0.015]). Averaged with boosting (weight chosen on validation) it adds a small but significant amount: +0.008 [+0.005, +0.012] to LightGBM and +0.004 [+0.001, +0.007] to the final model, so its errors differ from those of the trees.



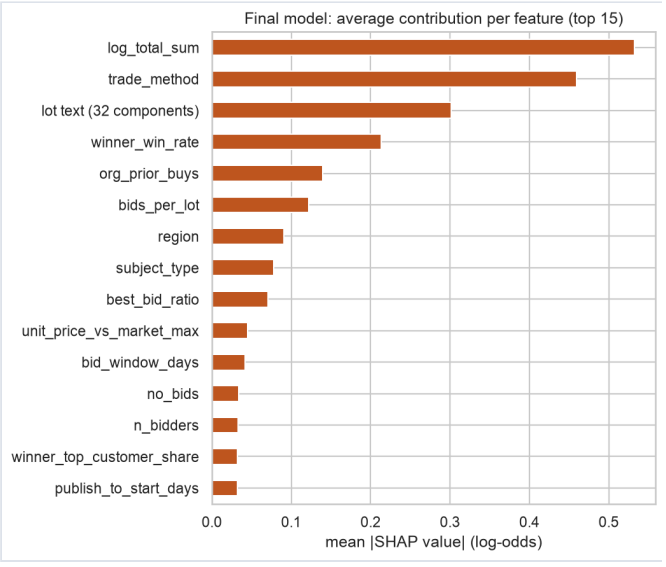


Figure 1. Left: test PR-AUC of every A3 model (dashed line: no model, 0.066). Right: mean absolute SHAP value of the final model's features (the 32 text components summed into one).

3.4 Final model, ablations and a leakage case

Model (test period, 38,981 procurements)	PR-AUC [95 % CI]	ROC-AUC	P@1%	R@5%
No model	0.066	0.500	0.057	0.044
Midterm baseline: logistic regression, 20 features	0.262 [0.246, 0.280]	0.792	0.504	0.263
Tuned CatBoost, 20 features	0.311 [0.293, 0.331]	0.807	0.573	0.307
Qwen2.5-1.5B + QLoRA	0.322 [0.302, 0.342]	0.808	0.594	0.320
Final: CatBoost, 61 features, calibrated	**0.364 [0.343, 0.385]**	**0.827**	**0.715**	**0.337**
Final + LLM ensemble	0.368 [0.347, 0.390]	0.826	0.715	0.342

The final model was chosen on validation among three candidates (CatBoost 0.405, simple LightGBM 0.400, LightGBM re-tuned on the 61 features 0.386 — tuning again did not help). For an auditor checking 1 % of procurements (≈ 390 in the test fortnight) it finds **≈ 280 failed contracts instead of ≈ 196** with the midterm model. The gain splits into two similar steps: model family (+0.049) and new information with the same CatBoost (+0.053). Ablations show that the information is almost entirely the **lot text** (−0.025 [−0.034, −0.016] without it), while the customer–supplier history and price features, predictive on their own, add nothing once the text is present — consistent with the finding that corruption red flags are almost uncorrelated with execution outcomes (Decarolis & Giorgiantonio, 2022). SHAP ranks value (0.53), method (0.46) and lot text (0.30) first, followed by the winner's past win rate (0.21). The largest improvements are for single-source purchases after a failed tender (lift over the base rate 2.2 → 8.0) and for the cheapest quarter of procurements (2.3 → 3.6), the midterm model's weakest segment.

Leakage found and removed. An earlier version of the research added "winner activity" features that looked like the third most important source of information (+0.025). The winner had been taken from the contracts table; inspecting the top of the ranked list showed that the contract signer had not even bid in 27 % of successful but 63 % of failed procurements — after a refusal the contract goes to someone else, so *who signed* is partly the outcome. Defining the winner from the announcement or the lowest bid (with the procedural tie-break) removes the gain entirely (−0.003 [−0.008, +0.002]). The leak was in the *link* between records, not in any single field — a case of the leakage patterns described by Kaufman et al. (2012).

4. Hypotheses

Hypothesis (written before the experiments)	Result
H4 The model family matters more than tuning	Supported: family +0.04–0.05; tuning 0 to +0.04, large only for poor defaults
H5 New information matters more than tuning	Supported: same CatBoost, +0.053 from information vs +0.023 from tuning
H6 Text is useful with the table, not alone	Supported: text alone 0.156 at best; the largest addition to the table (+0.036)
H7 Fine-tuning makes text a stronger signal than TF-IDF	Partly: a tie overall, a clear gain on unseen texts; frozen components are the better feature
H8 The LLM is below boosting but helps in an ensemble	Supported: −0.028 vs full-data boosting; +0.004 [+0.001, +0.007] in the ensemble
H9 Calibration fixes probabilities without changing the ranking	Supported: Brier 0.200 → 0.055 for the weighted baseline; PR-AUC unchanged

5. Limitations

All models are trained on January–February 2024 and tested on the second half of March: end-of-year procurement, when budgets are rushed, may fail more often and for other reasons. The label is the final status on the portal, and terminations "by mutual agreement" include harmless cases (a strict label is kept in the data). Only information known at award time is used, which by design excludes the strongest signals of the rule-based prototype (contract execution, payments, fines): they are the outcome itself. GPU results (fine-tuning, LLM) can differ in the third decimal across hardware.

6. Conclusions

A better model and better information contribute about equally: gradient boosting and new information each add about 0.05 PR-AUC, while hyper-parameter tuning beyond reasonable defaults adds little or nothing. The

most valuable new information is the text of what is being bought, and the best way to give it to a tabular model is a compact frozen embedding rather than a fine-tuned score; fine-tuning pays off when texts are new. A 1.5 B-parameter LLM tuned with QLoRA on a free GPU learns the task but does not beat boosting on hundreds of thousands of rows; it is useful as a second opinion in an ensemble. The final model ranks failed contracts with PR-AUC 0.364 and calibrated probabilities, finding about 40 % more failures than the midterm model at the same audit budget. Methodologically, the main lesson is that a suspiciously large gain must be traced to the moment each value — and each link between records — becomes known.

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